

TINKER AFB, OK, USA

WMO: 723540

Lat: 35.417N

Lon: 97.383W

Elev: 1291

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	18.4	2.9	6.7	18.5	7.4	8.4	24.2	30.2	42.6	27.8	42.4	11.5	0	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	100.0	73.4	97.0	73.7	93.8	74.0	78.2	90.2	77.1	89.2	76.1	88.1	11.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	137.7	83.5	73.5	130.6	82.3	72.7	127.3	81.6	42.6	90.2	41.4	89.1	40.3	88.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	24.3	21.3	7.1	102.9	3.7	4.0	4.4	105.8	2.3	108.2	0.2	110.4	-2.5	113.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	39.1	43.3	52.0	60.4	69.1	77.5	82.1	81.6	74.0	62.5	50.9	41.2
	DBStd	17.36	10.61	11.57	10.71	8.44	7.07	5.46	5.03	5.25	7.25	8.51	9.31	9.78
	HDD50	1135	360	242	103	17	1	0	0	0	0	11	100	301
	HDD65	3382	802	609	416	182	43	1	0	0	14	148	429	738
	CDD50	5235	24	55	165	328	592	825	996	978	721	397	126	28
	CDD65	2006	0	2	13	43	169	376	531	513	285	69	5	0
	CDH74	20907	2	20	117	352	1346	3710	6299	5861	2637	523	39	1
	CDH80	9720	0	4	21	68	392	1595	3316	3060	1134	127	3	0
Wind	WSAvg	10.8	11.6	11.7	12.5	12.8	11.0	10.3	9.3	8.7	9.0	10.5	11.4	11.0
Precipitation	PrecAvg	36.1	1.3	1.6	2.6	3.7	5.4	4.7	3.7	3.0	3.5	2.9	2.0	1.7
	PrecMax	53.3	3.3	4.1	6.5	7.1	16.2	11.3	8.7	7.7	8.9	10.2	6.1	5.7
	PrecMin	23.9	0.0	0.0	0.4	0.7	1.5	1.0	0.6	0.0	0.8	0.4	0.0	0.0
	PrecStd	7.6	0.9	1.1	1.7	1.7	3.4	2.6	2.3	1.9	1.8	2.1	1.8	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	76.6	82.8	86.1	91.2	99.5	104.1	104.7	100.4	88.0	78.7	70.4
		MCWB	53.6	57.3	62.1	64.8	71.2	74.9	72.6	73.3	71.7	69.4	63.0	58.2
	2%	DB	65.5	70.7	77.1	81.8	87.5	94.8	100.2	99.9	93.5	83.6	73.5	65.3
		MCWB	53.1	55.4	60.6	64.8	71.5	74.6	73.5	73.4	72.2	67.4	60.2	56.0
	5%	DB	60.7	65.7	72.7	78.3	84.3	91.2	97.0	97.0	90.3	80.0	69.6	60.8
		MCWB	50.3	54.3	58.7	62.9	70.1	75.0	74.0	73.7	72.0	65.4	58.9	51.4
	10%	DB	55.6	61.2	68.7	74.6	81.4	88.5	93.7	93.4	86.5	76.3	65.7	56.6
		MCWB	46.7	52.0	57.1	61.5	69.2	74.2	74.7	73.6	70.8	64.2	57.1	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.8	63.4	67.0	71.4	76.5	79.5	79.9	79.3	76.3	73.2	67.2	63.6
		MCDB	64.6	68.1	75.1	79.2	86.1	90.5	91.9	91.7	88.0	81.5	72.4	67.7
	2%	WB	56.1	60.4	64.4	68.7	74.4	77.5	78.2	77.6	74.8	71.2	63.9	58.5
		MCDB	61.2	65.7	72.3	75.7	82.9	88.6	90.5	90.5	86.3	78.7	69.4	61.9
	5%	WB	51.5	56.8	62.2	67.0	72.8	76.4	77.2	76.5	73.6	69.2	61.5	54.0
		MCDB	59.1	63.2	69.2	73.7	80.5	87.3	89.4	89.5	84.8	75.0	66.5	58.1
	10%	WB	47.4	52.3	59.3	64.7	71.1	75.2	76.1	75.4	72.7	67.1	58.9	49.1
		MCDB	54.8	61.2	66.8	71.6	78.6	85.7	88.3	88.4	83.2	73.1	64.3	55.8
Mean Daily Temperature Range	5% DB	MDBR	20.2	20.4	21.1	20.7	18.6	18.0	19.9	20.3	19.7	19.9	19.8	18.6
		MCDBR	27.7	27.1	26.3	24.8	21.3	21.1	24.4	25.2	23.4	23.6	24.6	24.8
		MCWBR	16.7	16.0	13.9	12.9	9.8	8.1	6.7	7.1	7.9	11.4	14.0	15.6
	5% WB	MCDBR	24.3	22.7	20.8	20.0	18.4	18.6	20.0	20.1	19.3	17.8	20.1	20.6
	MCWBR	16.8	16.6	12.7	12.8	10.2	8.7	8.1	8.0	7.7	10.8	14.5	16.8	
Clear-Sky Solar Irradiance	taub	0.294	0.298	0.339	0.375	0.412	0.421	0.423	0.432	0.385	0.344	0.312	0.290	
	taud	2.507	2.512	2.421	2.333	2.299	2.315	2.341	2.257	2.370	2.478	2.506	2.525	
	Ebn at Noon	290	302	296	288	278	274	272	268	276	280	280	285	
	Edh at Noon	26	29	35	40	42	41	40	43	36	30	26	24	
All-Sky Solar Radiation	RadAvg	878	1066	1395	1727	1864	2131	2158	1962	1622	1233	925	740	
	RadStd	88	162	88	128	200	206	115	121	144	158	121	84	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page